



National Technical University of Athens
School of Electrical and Computer Engineering
Doctor of Philosophy Program

Postgraduate Course:

103: Probabilistic Analysis of Electrical Systems

Wasserstein-Based Control Architecture:
From agent dynamics to optimal
transport regulation

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Abstract

Wasserstein distance is a metric defined on a metric space that measures the distance between two probability distributions, either discrete or continuous. It is commonly characterized through the optimal transport formulation, which provides a principled way to compute the minimum cost required to transform one distribution into another.

In this work, the Wasserstein distance is employed for the optimal navigation of a swarm of mobile robots. The swarm configuration is described as a probability distribution characterized by its mean and covariance. A Wasserstein-based control strategy is developed to regulate these statistical quantities toward prescribed desired values, ensuring that the collective behavior of the agents converges to a target Gaussian distribution in the optimal transport sense. The resulting control input is then translated into linear and angular velocity commands for each individual agent, while respecting the nonholonomic dynamics of wheeled mobile robots. This approach enables cooperative navigation of the swarm toward a common objective in an optimal manner.

1 Wasserstein-Based Distributional Control Framework

1.1 System Model

Consider a swarm of N wheeled mobile robots evolving in the plane. The position of agent i is denoted by

$$p_i(t) = [x_i(t) \ y_i(t)]^\top \in R^2, \quad i = 1, \dots, N.$$

Each agent is modeled by the unicycle dynamics

$$\begin{aligned} \dot{x}_i &= v_i \cos \theta_i, \\ \dot{y}_i &= v_i \sin \theta_i, \\ \dot{\theta}_i &= \omega_i, \end{aligned} \tag{1.1}$$

where v_i and ω_i denote the linear and angular velocities, respectively.

1.2 Probabilistic Representation of the Swarm

The collective configuration of the swarm is described by the empirical probability measure

$$\rho^N(t) = \frac{1}{N} \sum_{i=1}^N \delta_{p_i(t)}, \tag{1.2}$$

where $\delta_{p_i(t)}$ denotes the Dirac measure centered at $p_i(t)$.

We assume that, for sufficiently large N , the empirical measure converges weakly to a probability measure

$$\rho(t) \in \mathcal{P}_2(R^2),$$

where $\mathcal{P}_2(R^2)$ denotes the space of probability measures with finite second moment.

1.3 Wasserstein Distance

Let $(R^2, \|\cdot\|)$ be a Polish metric space. The Wasserstein-2 distance between two probability measures $\rho_1, \rho_2 \in \mathcal{P}_2(R^2)$ is defined as

$$W_2^2(\rho_1, \rho_2) = \inf_{\gamma \in \Pi(\rho_1, \rho_2)} \int_{R^2 \times R^2} \|x - y\|^2 d\gamma(x, y), \tag{1.3}$$

where $\Pi(\rho_1, \rho_2)$ denotes the set of all couplings of ρ_1 and ρ_2 .

1.4 Gaussian Approximation

Assume that the swarm distribution is well approximated by a Gaussian measure

$$\rho(t) = \mathcal{N}(\mu(t), \Sigma(t)),$$

where $\mu(t) \in R^2$ and $\Sigma(t) \in R^{2 \times 2}$ denote the mean and covariance, respectively. The desired distribution is given by

$$\rho_d = \mathcal{N}(\mu_d, \Sigma_d).$$

Under the assumption that $\Sigma(t)$ and Σ_d are diagonal, the squared Wasserstein-2 distance admits the closed-form

$$W_2^2(\rho(t), \rho_d) = \|\mu(t) - \mu_d\|^2 + \|\sigma(t) - \sigma_d\|^2, \tag{1.4}$$

where $\sigma(t)$ denotes the vector of standard deviations.

1.5 Distribution-Level Control Law

Define the distributional control input

$$u_d(t) = -k_m(\mu(t) - \mu_d) - k_s(\sigma(t) - \sigma_d), \quad (1.5)$$

with gains $k_m, k_s > 0$.

The empirical mean and variance are given by

$$\mu(t) = \frac{1}{N} \sum_{i=1}^N p_i(t), \quad \Sigma(t) = \frac{1}{N} \sum_{i=1}^N (p_i - \mu)(p_i - \mu)^\top. \quad (1.6)$$

1.6 Lyapunov Stability Analysis

1.6.1 Distribution-Level Lyapunov Function

Consider the Lyapunov candidate

$$V(t) = \frac{1}{2} W_2^2(\rho(t), \rho_d) = \frac{1}{2} (\|\mu(t) - \mu_d\|^2 + \|\sigma(t) - \sigma_d\|^2). \quad (1.7)$$

Clearly, $V(t)$ is positive definite and radially unbounded with respect to $(\mu - \mu_d, \sigma - \sigma_d)$.

1.6.2 Time Derivative of the Lyapunov Function

Taking the time derivative of $V(t)$ yields

$$\dot{V} = (\mu - \mu_d)^\top \dot{\mu} + (\sigma - \sigma_d)^\top \dot{\sigma}. \quad (1.8)$$

1.6.2.1 Mean dynamics From the definition of $\mu(t)$ and the agent dynamics, we obtain

$$\dot{\mu}(t) = \frac{1}{N} \sum_{i=1}^N \dot{p}_i(t).$$

Assuming velocity alignment with the distributional control field, we approximate

$$\dot{\mu}(t) = u_d(t).$$

1.6.2.2 Variance dynamics Under symmetric velocity fields and bounded heading errors, the variance dynamics satisfy

$$\dot{\sigma}(t) = -k_s(\sigma(t) - \sigma_d).$$

Substituting into $\dot{V}$ yields

$$\dot{V} = -k_m \|\mu - \mu_d\|^2 - k_s \|\sigma - \sigma_d\|^2 \leq 0. \quad (1.9)$$

Theorem 1.1 (Distributional Stability). The equilibrium $\rho(t) = \rho_d$ is globally asymptotically stable in the Wasserstein-2 metric.

Proof. The Lyapunov function $V(t)$ is positive definite and satisfies $\dot{V}(t) \leq 0$. Moreover, $\dot{V}(t) = 0$ if and only if $\mu = \mu_d$ and $\sigma = \sigma_d$. By LaSalle's invariance principle, the system converges asymptotically to the desired distribution. $\square$

2 Agent-Level Stability Analysis

In this section, we analyze the stability of the individual agents under the proposed Wasserstein-based control law. The objective is to show that each agent asymptotically aligns its velocity with the distribution-level control signal, ensuring physical realizability of the distributional control objective.

2.1 Agent Dynamics

Each agent is modeled as a unicycle-type wheeled mobile robot:

$$\begin{aligned}\dot{x}_i &= v_i \cos \theta_i, \\ \dot{y}_i &= v_i \sin \theta_i, \\ \dot{\theta}_i &= \omega_i,\end{aligned}\tag{2.1}$$

where $(x_i, y_i) \in R^2$ denotes the position of agent i , $\theta_i \in R$ its heading angle, and v_i, ω_i are the linear and angular control inputs, respectively.

2.2 Distribution-Level Reference Velocity

The distributional controller generates a reference velocity vector

$$u_d(t) = \begin{bmatrix} u_{d,x}(t) \\ u_{d,y}(t) \end{bmatrix} \in R^2,\tag{2.2}$$

which is assumed to be continuous and bounded:

$$\|u_d(t)\| \leq \bar{u} < \infty.\tag{2.3}$$

Define the reference heading angle

$$\theta_{\text{ref}}(t) = \text{atan2}(u_{d,y}(t), u_{d,x}(t)).\tag{2.4}$$

2.3 Control Law

Each agent applies the following control law:

$$\begin{aligned}v_i &= k_v \|u_d\| \cos(\theta_i - \theta_{\text{ref}}), \\ \omega_i &= -k_\omega \sin(\theta_i - \theta_{\text{ref}}),\end{aligned}\tag{2.5}$$

where $k_v > 0$ and $k_\omega > 0$ are control gains.

2.4 Error Coordinates

Define the heading alignment error

$$e_{\theta,i} := \theta_i - \theta_{\text{ref}}.\tag{2.6}$$

Define also the velocity tracking error

$$e_{v,i} := \begin{bmatrix} \dot{x}_i \\ \dot{y}_i \end{bmatrix} - u_d.\tag{2.7}$$

2.5 Heading Error Dynamics

Taking the time derivative of (2.6) yields

$$\dot{e}_{\theta,i} = \dot{\theta}_i - \dot{\theta}_{\text{ref}} = -k_\omega \sin(e_{\theta,i}) - \dot{\theta}_{\text{ref}}. \quad (2.8)$$

2.6 Assumption on Reference Signal

Assumption 1. The reference heading satisfies

$$|\dot{\theta}_{\text{ref}}(t)| \leq \bar{\omega}_d,$$

where $\bar{\omega}_d < k_\omega$.

This assumption is reasonable since $u_d(t)$ is generated by smooth distribution-level dynamics.

2.7 Lyapunov Analysis for Heading Dynamics

Consider the Lyapunov function candidate

$$V_{\theta,i} = 1 - \cos(e_{\theta,i}). \quad (2.9)$$

This function is positive definite with respect to $e_{\theta,i}$ and radially unbounded on $(-\pi, \pi)$. Taking the time derivative of $V_{\theta,i}$ along (2.8) yields

$$\begin{aligned} \dot{V}_{\theta,i} &= \sin(e_{\theta,i}) \dot{e}_{\theta,i} \\ &= -k_\omega \sin^2(e_{\theta,i}) - \sin(e_{\theta,i}) \dot{\theta}_{\text{ref}}. \end{aligned} \quad (2.10)$$

Using Assumption 1 and the inequality $|\sin(e)| \leq 1$, we obtain

$$\dot{V}_{\theta,i} \leq -(k_\omega - \bar{\omega}_d) \sin^2(e_{\theta,i}). \quad (2.11)$$

Theorem 2.1. Under Assumption 1, the heading error $e_{\theta,i}$ converges asymptotically to zero.

Proof. The Lyapunov function $V_{\theta,i}$ is positive definite and satisfies $\dot{V}_{\theta,i} \leq 0$, with $\dot{V}_{\theta,i} = 0$ if and only if $e_{\theta,i} = 0$. By LaSalle's invariance principle, $e_{\theta,i}(t) \rightarrow 0$ as $t \rightarrow \infty$. $\square$

2.8 Velocity Tracking Analysis

From (2.1) and (2.5), the velocity of agent i satisfies

$$\begin{bmatrix} \dot{x}_i \\ \dot{y}_i \end{bmatrix} = k_v \|u_d\| \begin{bmatrix} \cos \theta_i \\ \sin \theta_i \end{bmatrix} \cos(e_{\theta,i}). \quad (2.12)$$

As $e_{\theta,i} \rightarrow 0$, we have

$$\cos(e_{\theta,i}) \rightarrow 1, \quad \begin{bmatrix} \cos \theta_i \\ \sin \theta_i \end{bmatrix} \rightarrow \frac{u_d}{\|u_d\|}.$$

Thus,

$$\lim_{t \rightarrow \infty} \begin{bmatrix} \dot{x}_i \\ \dot{y}_i \end{bmatrix} = k_v u_d. \quad (2.13)$$

Choosing $k_v = 1$ yields exact velocity tracking.

2.9 Agent-Level Stability Result

Theorem 2.2. Under the proposed control law, each agent asymptotically aligns its velocity with the distribution-level control signal $u_d(t)$.

Proof. The heading error converges asymptotically to zero by Theorem 2.1. Consequently, the agent velocity converges to $u_d(t)$. This establishes asymptotic velocity alignment for each agent. $\square$

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